Optimal operation of the paratransit in small urban area during COVID-19 pandemic: risk assessment and exposure mitigation

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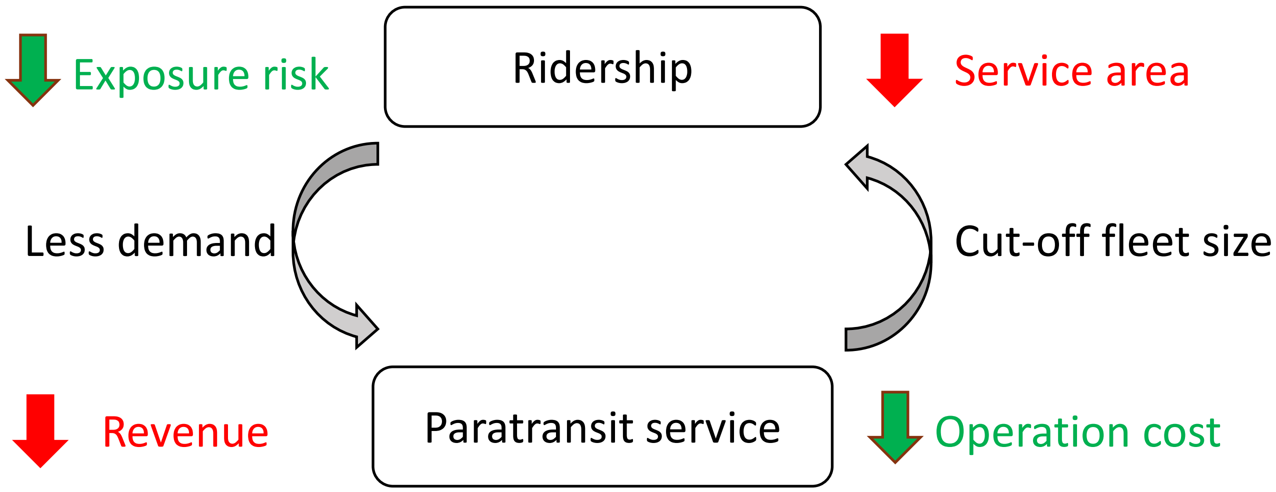
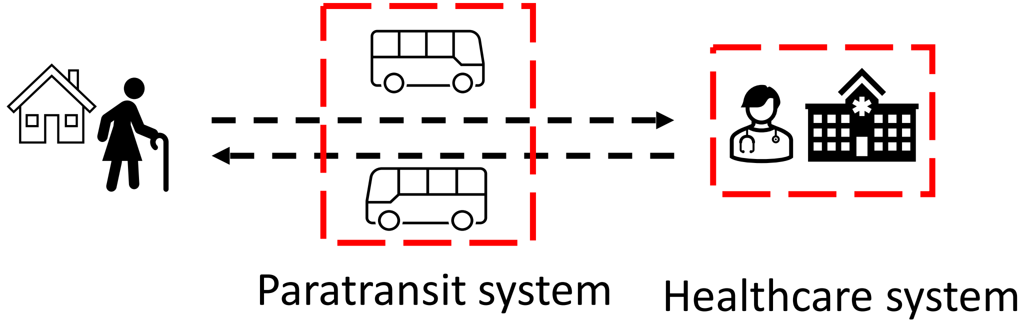
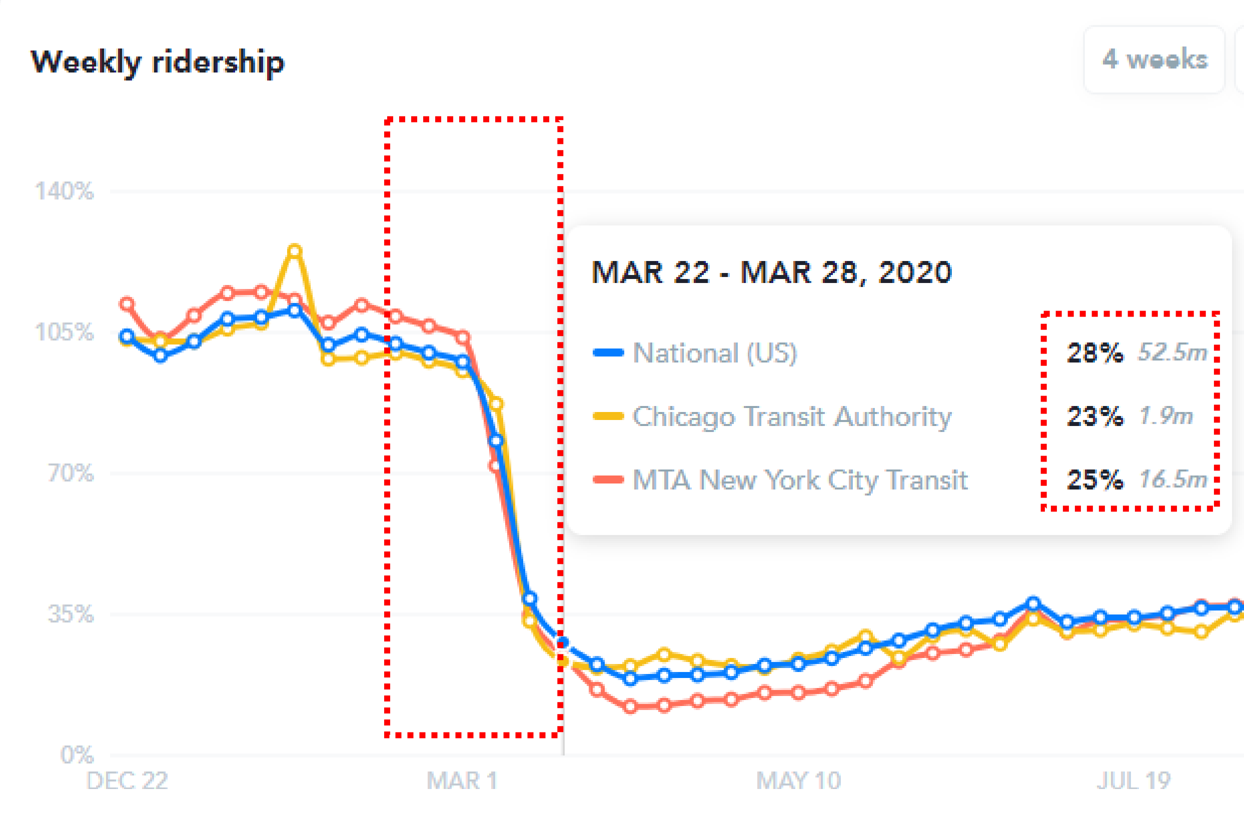
Background

After the COVID-19 pandemic:



For people with disability and pre-existing conditions:





Weekly ridership of public transit, source: transitapp.com

Research questions

We focus on the designing a ***risk-aware*** operation for paratransit services.

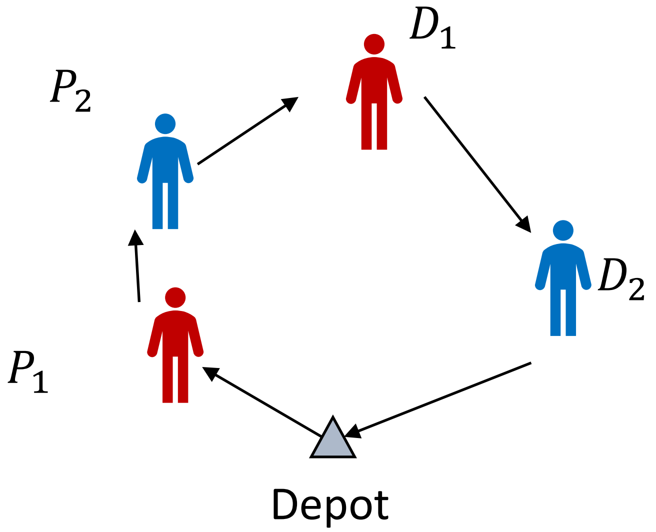
- Can we provide an ***efficient disease-resilience*** routing strategy with ***joint consideration*** of exposure risk and operation cost?
- What is the ***level of service*** and ***operation cost*** after implementing the risk-aware operation?
- What is the ***trade-off*** between the level of risk and operation cost from ***different perspectives*** of stakeholders?

Methodology

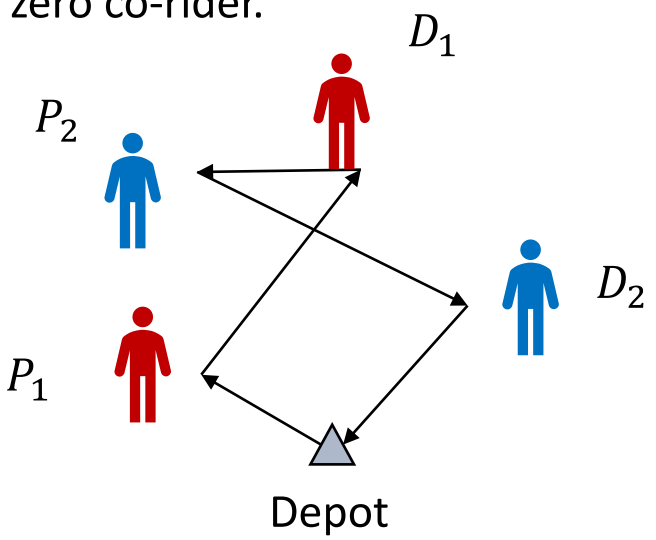
Dial-a-ride problem (DARP) [1]:

- Complete directed graph, $G = (N, A)$.
- Four subsets of nodes: $N = \{0\} \cup \{2n + 1\} \cup P \cup D$.
- Time window $[e_i, l_i]$ for each node i.
- Number of passengers $\{1, -1\}$; risk level $\{r_i, -r_i\}$.

Route 1: one co-rider.



Route 2: zero co-rider.



[1] Ropke, S., Cordeau, J. F., & Laporte, G. (2007). Models and branch-and-cut algorithms for pickup and delivery problems with time windows. *Networks: An International Journal*, *49*(4), 258-272.

Methodology

Risk-aware routing strategy:

$$\min_{x}\left(w_1 \sum_{(i,j)\in\mathscr{A}} t_{ij}x_{ij} + w_2\ |\mathscr{P}|\ \max_{i\in\mathscr{P}}(H_i)\right)$$

Minimize the total cost and the maximum individual risk

s.t.

$$\sum_{i\in\mathscr{N}} x_{ij} = 1$$
$$\sum_{j\in\mathscr{N}} x_{ij} = 1$$

Flow conservation constraints.

$$\sum_{i\in S}\sum_{j\in S} x_{ij} \le |S| - 2$$

Precedence constraints [2]

$$\Gamma_j \ge \Gamma_i + t_{ij} - B\left(1 - x_{ij}\right)$$
$$e_i \le \Gamma_i \le l_i$$
$$\Gamma_i \le e_i + t_{i,n+i} + \Delta t_i$$

Time window constraints.

$$C_j \le C_i + c_j - B(1 - x_{ij})$$
$$R_j \le R_i + r_j - B(1 - x_{ij})$$
$$\max\{0, c_i\} \le C_i \le \min\{C, C + c_i\}$$

Consistency of on-board passenger number and cumulative risk level

$$Q_j \ge Q_i + q_{ij} - B\left(1 - x_{ij}\right)$$
$$q_{ij} = t_{ij}R_i$$

Propagation of on-route risk level

$$H_i = Q_{n+i} - Q_i - (\Gamma_{n+i} - \Gamma_i)\, r_i$$

Individual risk level

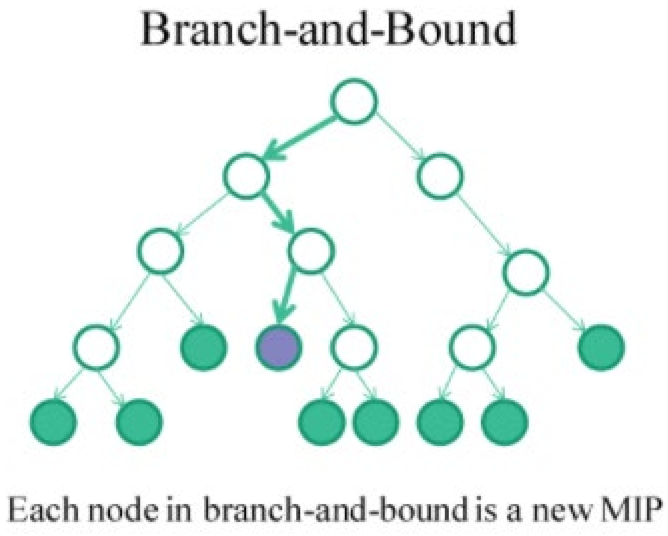
$$x_{ij} \in \{0, 1\}$$

Binary decision variable

[2] Cordeau, J. F. (2006). A branch-and-cut algorithm for the dial-a-ride problem. *Operations Research*, *54*(3), 573-586.

Methodology

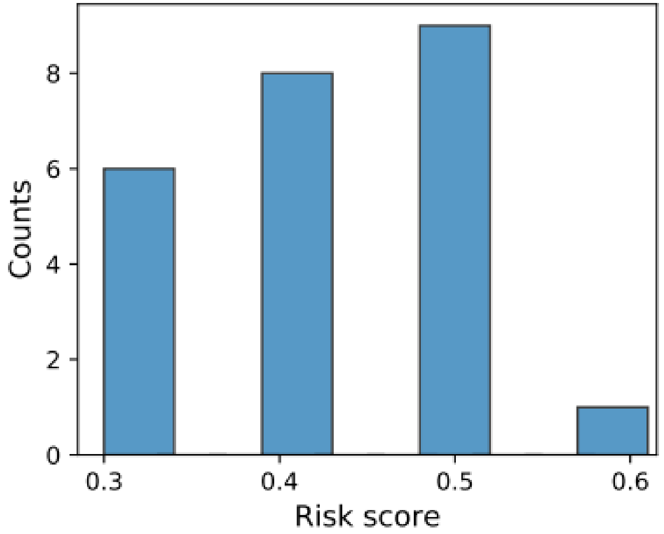
- Branch-and-cut algorithm
  - Precedence constraints
  - Subtour elimination
- Initial pool of inequalities [2]
  - Helps to tighten the constraints
- Arc elimination
  - Reduce the problem size
  - Triangle inequality w.r.t. time window constraints

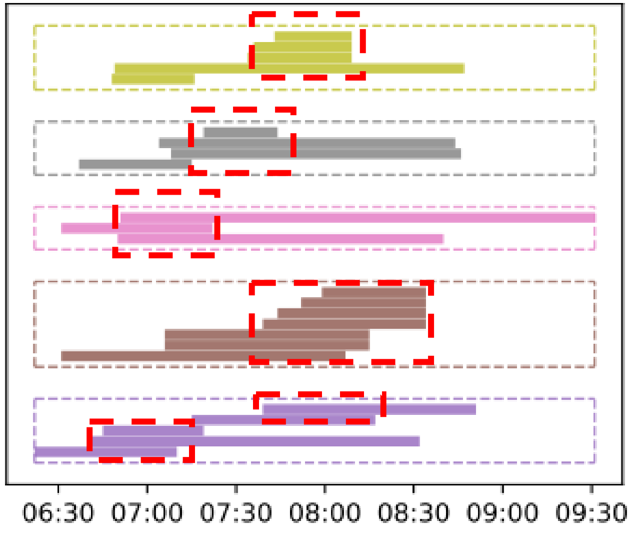


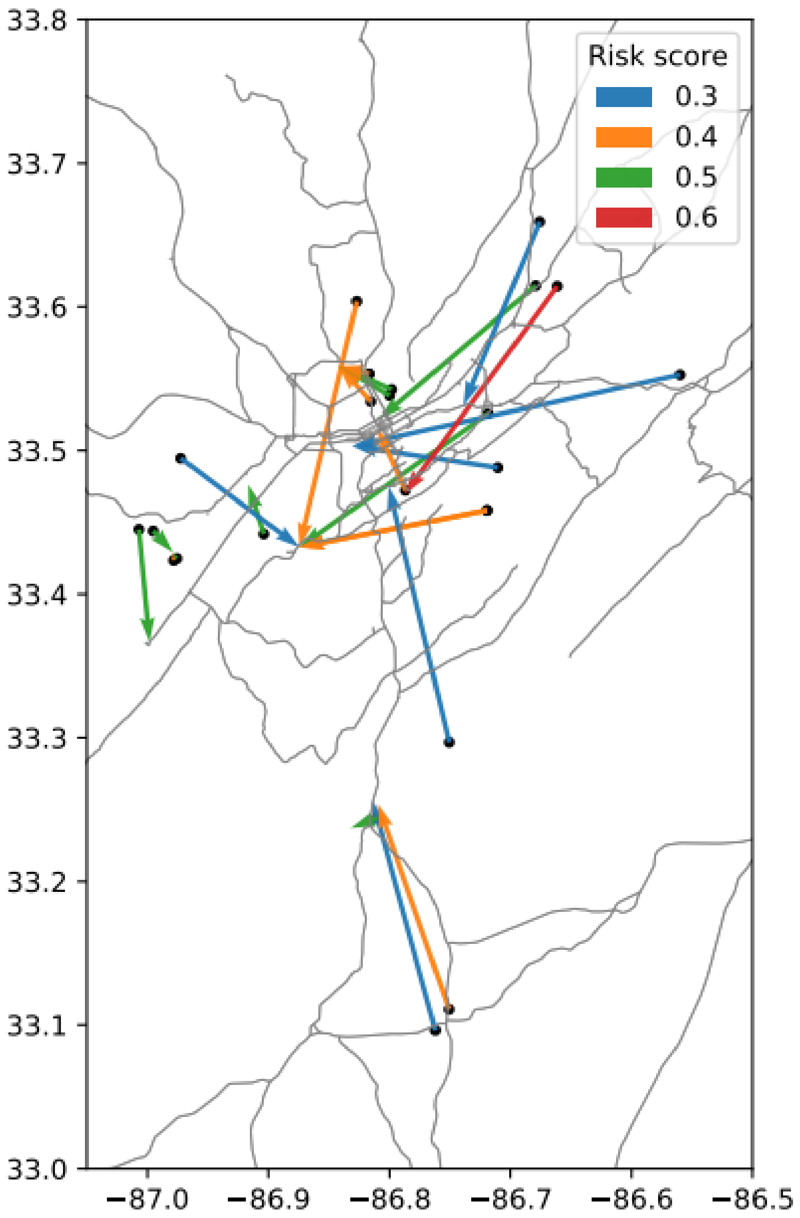
Sample of branch-and-cut algorithm, source: Gurobi

Case Study

- Paratransit service in central Birmingham area, AL.
- Day-ahead reservation
- 3/11/2020
- First 24 passengers, originally assigned to 5 shuttles.
- Risk level: travel purposes, ages, County of origin.







Details of selected passengers and trips

Results

- Observed trips: as in historical data
- Base-DARP: $w_1 = 20, w_2 = 0, l_i - e_i = 15\text{min}$
- Risk-DARP: $w_1 = 20, w_2 = 6, l_i - e_i = 15\text{min}$

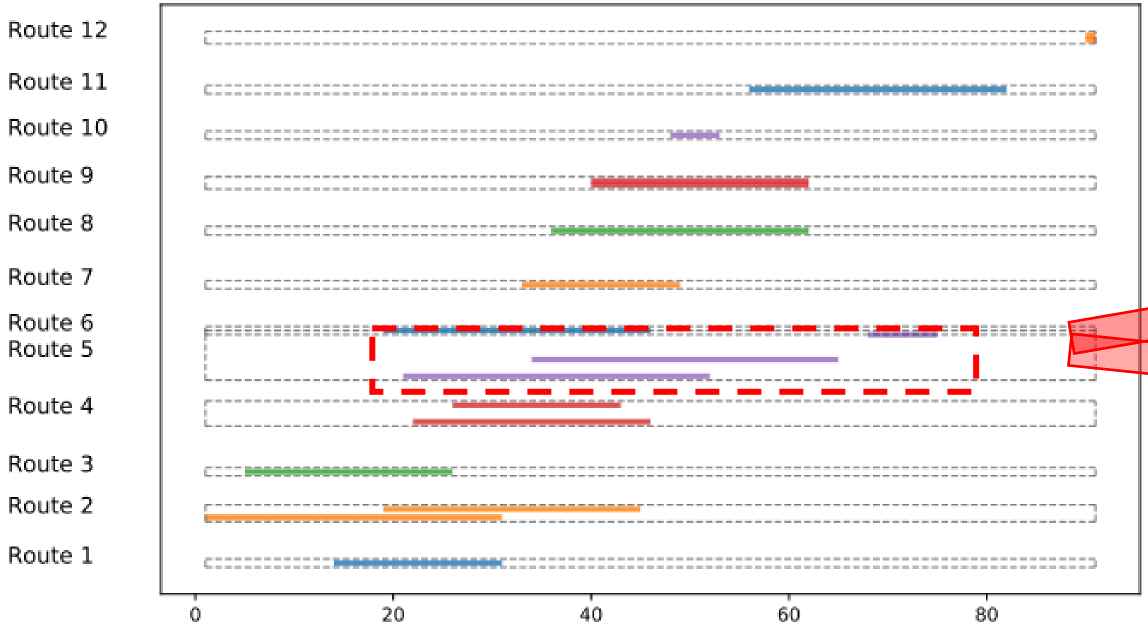
$$\min_{x} \left(\boxed{w_1} \sum_{(i,j)\in\mathscr{A}} t_{ij}x_{ij} + \boxed{w_2} \, |\mathscr{P}| \max_{i\in\mathscr{P}} (H_i) \right)$$

- # of co-riders
- Total travel time
- Maximum individual risk

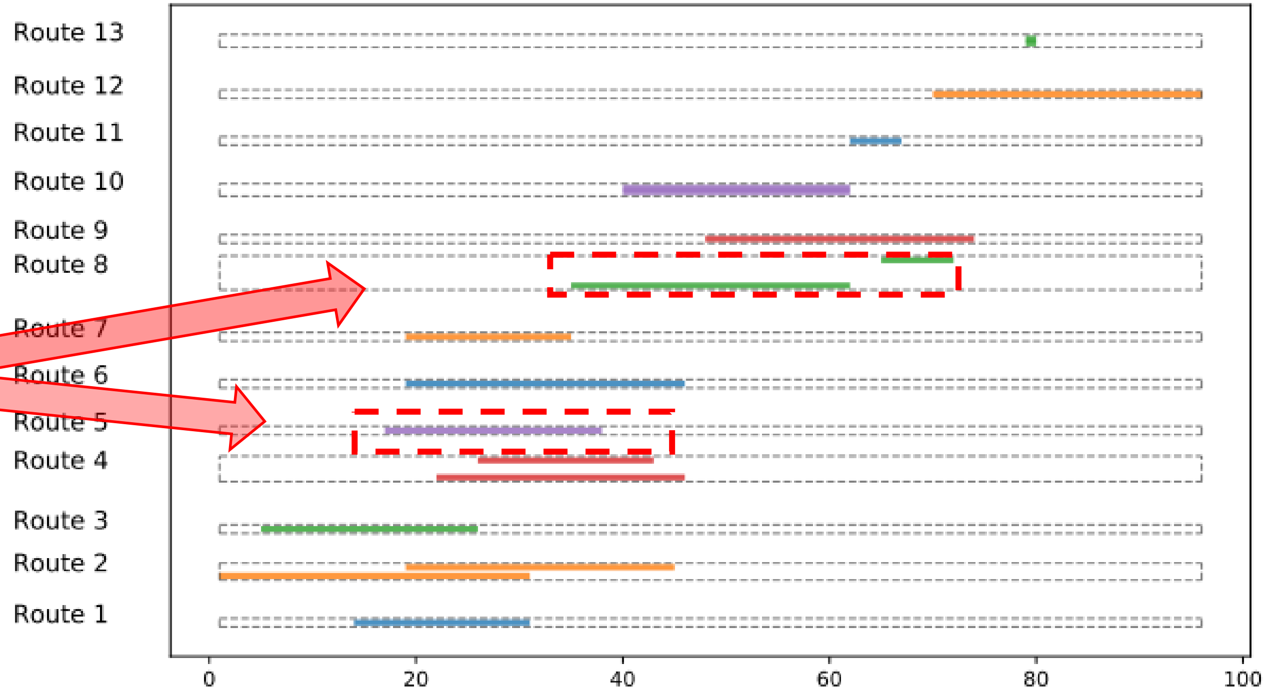
$\|\mathscr{P}\|$	Waiting time (min)	Trip delay (min)	Number of shuttles (or routes)	Max number of co-riders	Total travel time	Max individual risk	CPU Time (sec)	MIP Gap (%)
Observed Trips								
6	7.96	13.39	2	2	245	52.8	-	-
8	9.01	16.11	2	2	253	52.8	-	-
10	8	13.1	2	4	272	100.4	-	-
12	8.5	12.79	2	6	272	120.1	-	-
14	8.75	12.54	3	6	401	120.1	-	-
16	9.64	11.74	4	6	490	120.1	-	-
18	9.64	10.28	4	6	581	120.1	-	-
20	8.69	9.79	5	6	609	120.1	-	-
22	8.24	9.31	5	6	700	120.1	-	-
24	7.48	8.55	5	6	700	120.1	-	-
Base-DARP ($w_1 = 20,\ w_2 = 0,\ l_i - e_i = 15$)								
6	3.50	3.50	4	1	139	10.5	0.02	0
8	4.75	2.12	6	1	190	10.5	0.02	0
10	4.80	1.40	7	1	221	10.8	0.22	0
12	3.92	2.54	9	1	277	10.8	1.37	0
14	5.43	1.07	10	1	286	10.8	2.35	0
16	5.69	0.62	11	1	326	15.0	11.82	0
18	4.22	1.83	12	1	331	28.2	29.15	0
20	4.80	1.72	13	1	359	15.4	74.96	0
22	3.18	3.77	14	1	368	10.8	645.32	0
24	4.50	2.12	14	2	370	11.2	>1800	3
Risk-DARP ($w_1 = 20,\ w_2 = 6,\ l_i - e_i = 15$)								
6	2.33	2.83	5	1	142	6.0	0.1	0
8	3.50	3.50	8	0	205	0.0	0.11	0
10	2.90	1.75	8	1	229	6.8	0.3	0
12	3.58	3.00	10	1	285	6.8	1.6	0
14	3.07	3.50	11	1	294	8.8	6.16	0
16	4.31	1.97	12	1	334	8.8	126.41	0
18	4.39	1.75	13	1	339	8.8	349.37	0
20	4.40	2.10	15	1	367	8.8	>1800	11
22	3.23	3.30	16	1	376	8.8	>1800	16
24	3.79	2.75	15	2	370	10.8	>1800	19

Results

Detailed routing schedule



(a) Base-DARP

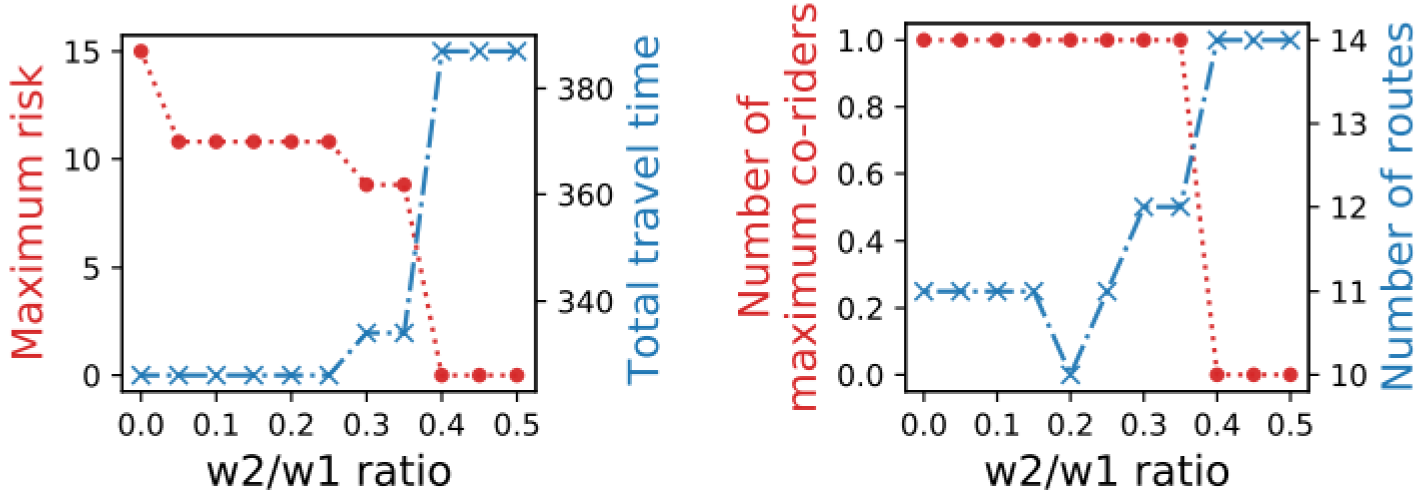


(b) Risk-DARP

Results

Sensitivity analysis:

- consideration on the risk mitigation (w2/w1 ratio)



- Passengers ($\frac{w2}{w1} = 0.4$): zero risk
- Paratransit operators ($\frac{w2}{w1} = 0.2$): low cost and reduced fleet size
- State agencies ($\frac{w2}{w1} \geq 0.4$): subsidize the operators for operation with larger fleet size.

Conclusion

- Our proposed cases, Base-DARP and Risk-DARP can both significantly ***reduce the risk level*** while ***ensuring low travel cost***.
- A trade-off point at ***w2/w1=0.4*** suggests a relatively ***balanced*** performance regarding the risk and travel cost.
- ***Subsidies*** in paratransit operator are required to further improve the performance on risk mitigation and travel time.

Thank you!

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